CLASS 7 — MATHS · FORMULA SHEET

Chapter 4: **Expressions using Letter-Numbers** | Quick revision, one page

Letter-numbers and expressions

- A letter stands for a number that can change (variable). $3n$ means $3 \times n$; $n \times n = n^2$.
- Term: a part of an expression joined by $+$ or $-$. Expression: terms combined, e.g. $3n + 5$.
- Like terms have the same letter part ($4x$ and $9x$); unlike terms have different letter parts ($4x$ and $3y$).

Forming expressions

- 'n more than a' $\rightarrow a + n$; 'n less than a' $\rightarrow a - n$; 'k times x' $\rightarrow kx$.
- Perimeter of a square $= 4s$; of a rectangle $= 2(l + b)$. Area of a rectangle $= lb$.
- Cost of n items at ₹p each $= np$. Fixed charge + rate $\times$ units: $a + bn$.

Evaluating

- Substitute the number for the letter and follow BODMAS: if $x = 4$, then $2x + 3 = 2 \times 4 + 3 = 11$.

Adding and subtracting

- Add or subtract only like terms: $3a + 2a = 5a$; $7x - 4x = 3x$.
- $4x + 3y$ cannot be simplified further.
- Subtracting: change the sign of every term being subtracted: $(5x + 3) - (2x - 4) = 3x + 7$.
- Brackets with a minus sign in front: $-(a - b) = -a + b$.

Patterns

- A rule such as $2n + 1$ gives the nth term; find n for a given term by solving, e.g. $2n + 1 = 41 \Rightarrow n = 20$.

MASTERMIND